

Work, Force, Energy & Simple Machines (Part - 2)

Force

A push or pull acting on an object is called a force.

A force can be used to:

- Move a stationary object
- Stop a moving object
- Make a moving object move faster
- Slowdown a moving object
- Change the direction of a moving object
- Change the shape and size of an object.

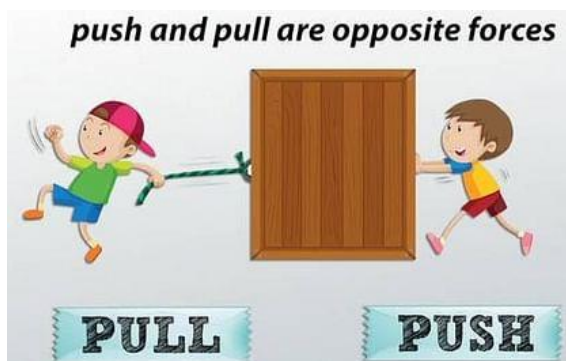


Fig: Force

Type of Force

There are different types of forces

Muscular Force

When something has to be pushed, pulled or lifted, muscular force is applied.



Fig: Muscular Force
Gravitational Force

Because of gravitational force, we are able to stay on the ground.

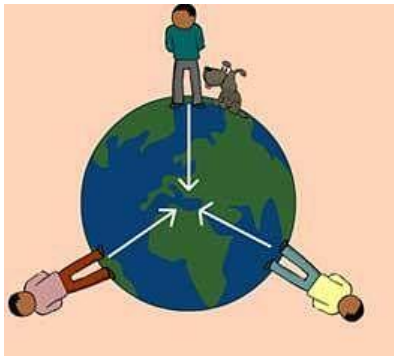


Fig: Gravitational Force

Frictional Force

Frictional force slows down a moving object. Even frictional force helps us in walking.

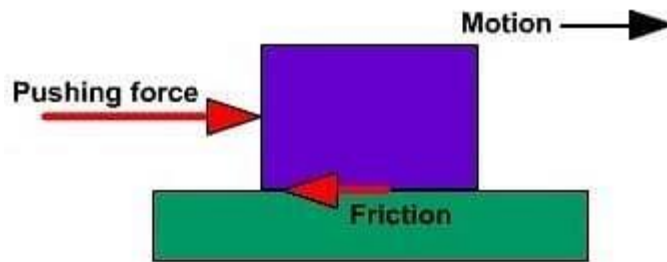


Fig: Frictional Force

Elastic Force

A stretched rubber band regains its original position on being released because of elastic force. Elastic force arises when a body deforms.



Fig: Elastic Force

Mechanical Force

Simple tools makes use of mechanical force.

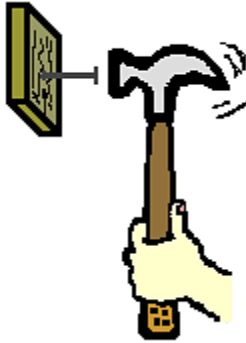


Fig: Mechanical Force

Buoyant Force

The upward push of water on a floating object is called buoyant force or up thrust. If we push a mug or piece of wood, we can feel an upward thrust.

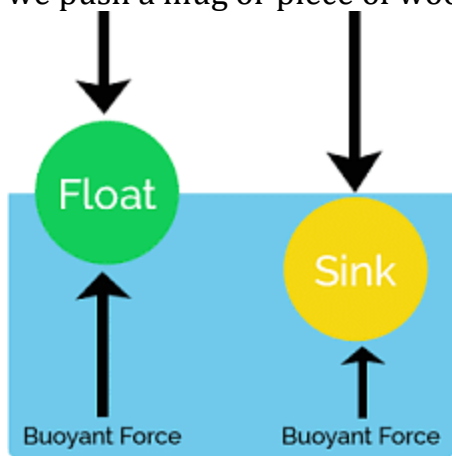


Fig: Buoyant Force

Energy:

We require energy to do work. We get lots of nutrients from plants and animals, which are the source of energy. These nutrients in our body is digested and give us energy. A small animal like ass cannot carry load equal to the load carried by an elephant. Hence, the energy is the capacity to do work. Every movement in our universe is carried out by the consumption of energy.

Therefore, we need different forms of energy to do various types of work. Heat energy is required to boil the water. Our voice and sound produced by the different objects possess sound energy. Mechanical energy is required to lift and or move an object. We use muscular energy to move our body. A bulb in our house works on

electrical energy. Mechanical energy is further divided into two groups, kinetic and potential energy.

Kinetic Energy

The energy that an object possesses by its motion is called kinetic energy. A stone thrown on the glass, breaks the glass due to kinetic energy. A ball hits the stamps due to the kinetic energy. Therefore, a moving object possesses kinetic energy.

Kinetic energy of a moving object is depends on the mass of the object. If the mass of the object is more, it possesses greater kinetic energy.

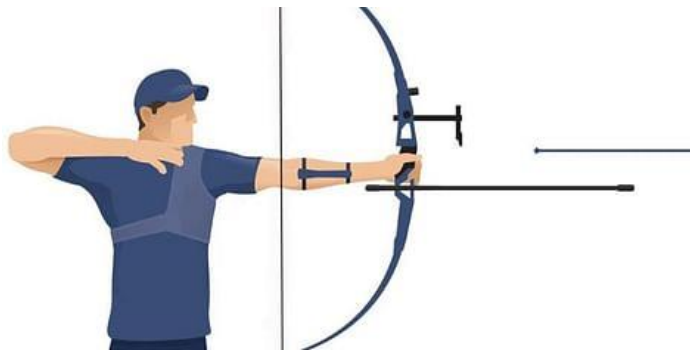


Fig: Kinetic Energy

If we throw an iron ball slowly on a soft piece of land, the ball will float on the surface. If thrown fast then the ball will dig the land and will enter into the lower surface. Hence, the kinetic energy of an object is dependent on the speed of the object.

Look at the following picture of examples of kinetic energy:

Image

Potential Energy

When a body is placed at certain height from the surface of the earth, the energy it possesses is called potential energy. The formula to calculate the potential energy is $m \times g \times h$. Where, m = mass of the body, g = gravitational pull by the earth and h = height of the body from the surface of the earth.

Therefore, the potential energy of a body depends on the mass and height at which the body is placed from the surface of the earth. The potential energy will increase on increasing the height from the surface of the earth and also the mass of the body.

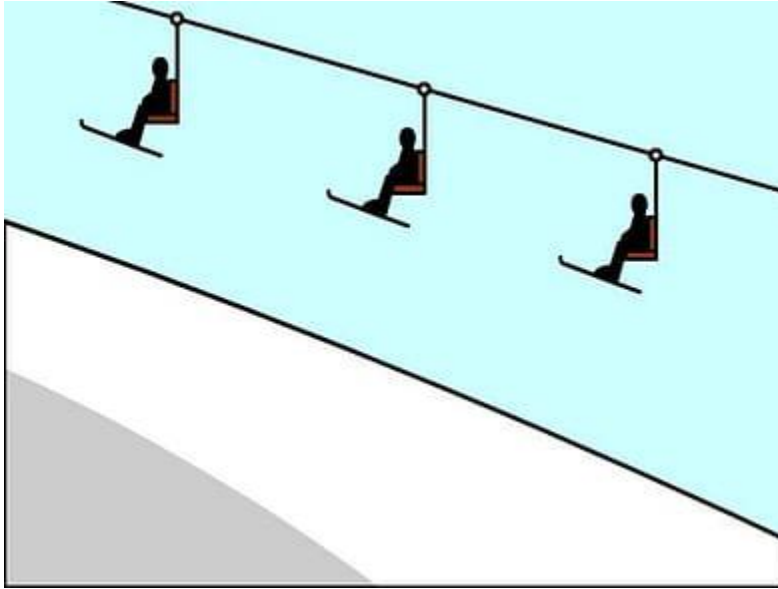


Fig: Potential Energy

Look at the following picture, the example of potential energy::

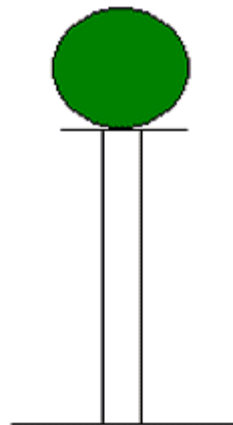


Fig: A Ball placed at some height 'h'

In the above picture a ball of mass m is placed at the height of h . The ball possesses potential energy and energy possessed by the ball is calculated by the formula: product of mass of the ball, gravitational pull or gravitational constant (9.8) and height from the surface of the earth. Hence the potential energy possessed by the ball = $m \times g \times h$.

Principle of Conservation of Energy

The energy possessed by a body is always remains constant and no energy is externally supplied and taken away from the body. But the energy stored in the body is transformed from one form of energy to another form. This principle of energy is known as the conservation of energy.

Sources of Energy

Sun is the primary source of energy for all living organism. Plants require sun energy for making food for themselves, which gives the energy to the plant for their internal activity. We know that all living organism directly or indirectly depend on the plants, hence the plants are the source of energy for rest of the living organism.

Wind and water are also the source of energy, which are used for the generation of electricity.

Depending upon the availability of different sources of energy, it can be classified into two types, renewable and nonrenewable source of energy.

Renewable Sources of Energy

The sources of energy which are constantly replenished by nature are called renewable sources of energy. For example. Sun, Wind and Water.

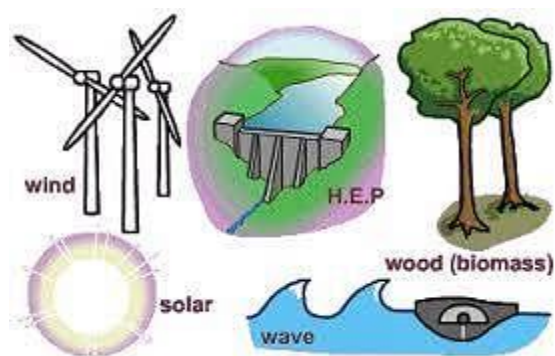


Fig: Renewable Sources of energy

Non - Renewable Sources of Energy

Some sources of energy require millions of years to be formed. These sources of energy are called non - renewable sources of energy. For example, fossil fuel.

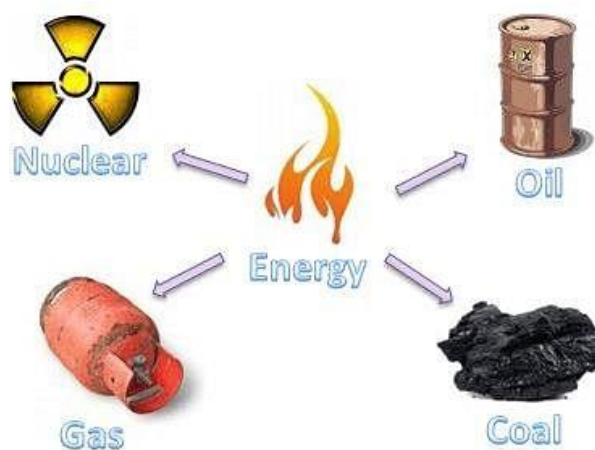


Fig: Non-renewable sources of energy

Simple Machines

Machines are simple tools which make our work easier and faster. They help us to do many work.

Some example of simple machines are lever, inclined plane, wheel and axle, pulley and screw.

Levers

Levers are commonly used tools like scissors, hammers and screw drivers.

When we lift a rock with the help of a rod, the weight lifted by the person is the load and the force with which it is lifted is the effort. The point of contact of rod and the stone is the fulcrum.

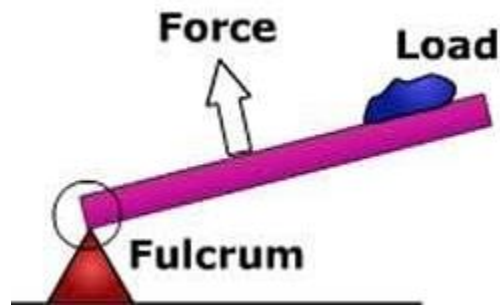


Fig: Lever

Levers can be classified according to the position of the fulcrum, the load and the effort.

When the fulcrum lies between the load and the effort, it is called first class lever.

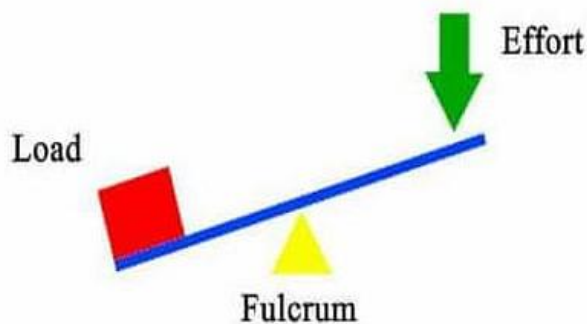


Fig: First Class Lever

When the load is in between the fulcrum and the effort, it is a second class lever.
When the effort is in between the fulcrum and the load, it is a third class lever.

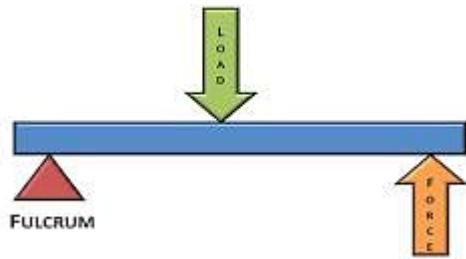


Fig: Second Class Lever

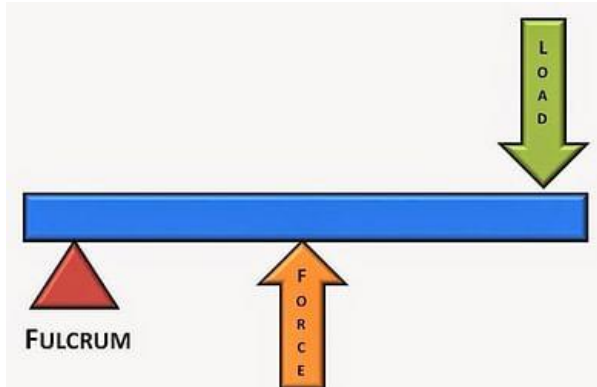


Fig: Third Class Lever

Inclined Plane

An inclined plane is a slope which makes work easier. When workers have to load or unload a truck they use a plane of wood as an inclined plane.



Fig: An Inclined Plane

The Pulley

A pulley is a small wheel with a groove around its outer edge. There are two types of pulleys a fixed pulley and a movable pulley. The pulley used for drawing water from a well is a fixed pulley movable pulleys along with fixed pulleys are used to lift loads.

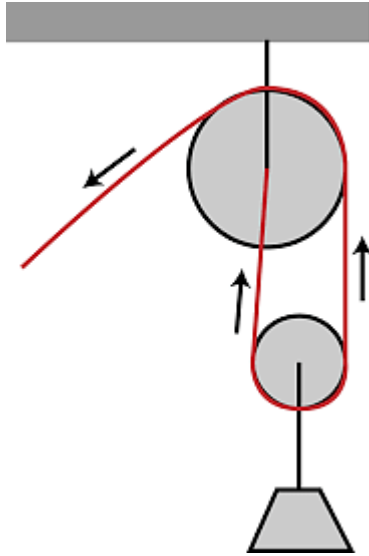


Fig: The Pulley

The wheel and axle

A wheel attached with a rod attached to it is known as a wheel-and-axle arrangement. It is easier to push a load on wheels than carrying it. This arrangement is used in vehicles, sewing machines and cycles.

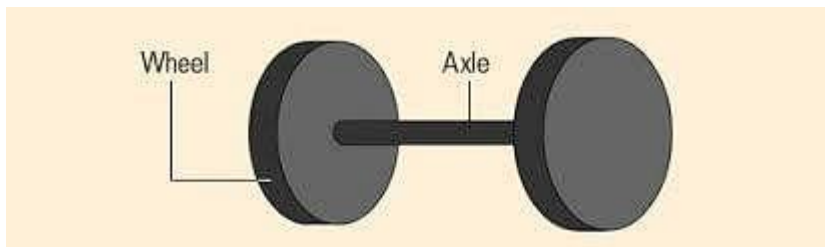


Fig: The wheel and the axle

The Screw

Screw is a tool used to hold things tightly together. Bigger screws called screw jacks are used to lift cars and other heavy objects. When two things are joined by a screw, they are held together through a long distance while when we join things with a nail, they are held together only through the length of the nail.



Fig: The Screw

Commonly Asked Question

1. Consider the following statements:

Statement: 1. Energy is the capacity to do work.

Statement: 2. Energy is the exerted force on a body.

Which one of the following is correct about the above statements?

- (a) Both statements are true
- (b) Both statements are false
- (c) Statement 1 is true and 2 is false
- (d) Statement 1 is false and 2 is true
- (e) None of these

Answer: (c)

Explanation

Energy is the capacity to do work. Therefore, option (C) is correct and rest of the options is incorrect.

2. Which one of the following types of energy is possessed by a body when placed at a certain height?

- (a) Kinetic energy
- (b) Potential energy
- (c) Heat energy
- (d) Sound energy
- (e) None of these

Answer: (b)

Explanation

When a body is placed at a certain height, body possesses potential energy. Therefore, option (B) is correct and rest of the options is incorrect.